

Koa Shen

B.S. Candidate in Mechanical Engineering · UC Santa Barbara · Robotics · Motion Systems

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EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering — 3.97 Major GPA · 3.94 Cumulative GPA

Santa Barbara, CA

- Relevant coursework: Statics, Dynamics, Dynamical Systems (in progress), Machine Learning (in progress), Intro and Advanced Mechanics of Materials, Circuits, MATLAB, and Calculus-based Physics.
- Consistently earned Dean's Honors in the College of Engineering.

EXPERIENCE

Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Worked across CAD import, kinematic modeling, collision checks, and mechanical load cases for high-DOF beamline assemblies.
- Applied model findings to the XCS optics redesign and documented interfaces across Solid Edge, Teamcenter, and the controls team's EPICS environment.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Serviced 100+ public exhibits and led three major overhauls; the Arp Forms redesign avoided more than \$3,000 in replacement costs.

Fabrication Team Lead

Fall 2024 – Present

Gaucha Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led a 15-person Chassis & Ergonomics team through GR26 safety-system design, manufacturing, and installation in 2025–26.
- CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum, holding critical bearing interfaces within 0.0005 in and standardizing five spindles around proven CAM.

SELECTED PROJECTS

Digital Twin Simulation Framework for LCLS Motion Stages

2026 – Present

- Built a Drake digital twin for an LCLS assembly (32 DOF, 305 bodies), using OpenCascade CAD import, CoACD collision meshes, and MeshCat; millisecond checks span 15 cataloged stages.
- Applied it to three assemblies; retroactive XCS checks exposed at least five pinch points restricting use.

Polycapillary Optics Assembly Redesign

2026 – Present

- Developed a redesign for a 32-DOF optics assembly with three kinematic bases and six hot-swappable optics; projected savings of 1–2 hours per shift-day and roughly \$30k in annual value are not yet verified.

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Designed a compact 15:1 cycloidal reducer within a NEMA 17's 42 mm square face for a planned 6-DOF arm; 381 mm reach and 0.5 kg payload are unvalidated design targets.
- Built PlatformIO C++ firmware for TMC2209 stepper control, including step/dir generation and real-time telemetry.
- Integrated a 12-bit AS5600 magnetic encoder to detect missed steps and characterize reducer motion; multi-joint output sensing remains planned.

FSAE Steering Wheel Development

Fall 2024 – Present

- Integrated buttons, potentiometers, a display, and quick-release hardware; reduced mass from 5 lb in 2024 to 1.7 lb in 2026 through iterative design and weight-optimization FEA.

TECHNICAL SKILLS

CAD & Manufacturing: SolidWorks, Solid Edge, Inventor / Fusion, Mastercam, GD&T, CNC / manual machining, Composites, Welding, Sheet metal

Programming: Python, C/C++, PlatformIO, MATLAB, Arduino / microcontrollers, Git / GitHub

Robotics & Controls: Drake kinematics (path planning in development), Collision detection, TMC2209 stepper control, AS5600 encoders / I2C multiplexing